INCH-POUND

MIL-DTL-32434/6
23 January 2013

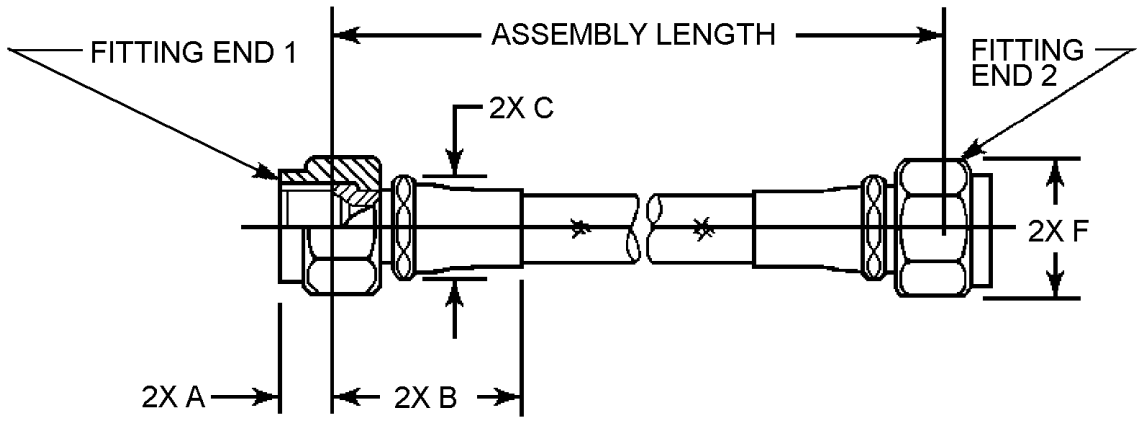
DETAIL SPECIFICATION SHEET

# HOSE ASSEMBLY, POLYTETRAFLUOROETHYLENE, CONVOLUTED, PERMANENTLY ATTACHED FITTINGS, HIGH TEMPERATURE, MEDIUM PRESSURE, FLARE-TO-FLARE

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32434.

Hose assembly dimensions for style A: See figure 1.




| Fitting ends | | Hose size | Dimensions | | | |
|---|---|---|---|---|---|---|
| 1 | 2 | | A (nom) | B ( max) | C (max) | F (max) |
| Straight | Straight | 3/4 | .372 | 1.049 | .620 | .580 |
| | | 4 | .375 | 1.049 | .620 | .654 |
| | | 5 | .380 | 1.194 | .680 | .726 |
| | | 6 | .380 | 1.303 | .750 | .798 |
| | | 8 | .438 | 1.465 | .880 | 1.014 |
| | | 10 | .515 | 1.680 | .967 | 1.158 |
| | | 12 | .570 | 1.727 | 1.122 | 1.447 |
| | | 16Z | .620 | 2.101 | 1.480 | 1.736 |
| | | 20Z | .640 | 2.255 | 1.750 | 2.328 |
| | | 24Z | .765 | 2.400 | 2.150 | 2.621 |

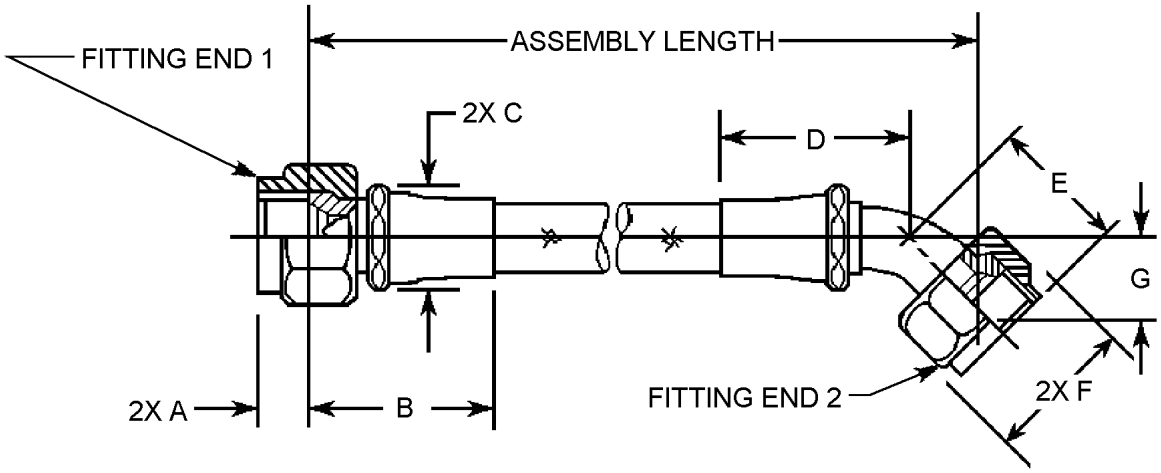
NOTE: Style A uses class 1 fittings which are made from corrosion resistant steel (CRES).

FIGURE 1. Style A hose assembly dimensions.



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Hose assembly dimensions for style B: See figure 2.




| Fitting ends | | Hose | Dimensions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | size | A (ref) | B (max) | C (max) | D (max) | E (max) | F (max) | G (min) | G (max) |
| Straight | 45° | 3/4 | .372 | 1.049 | .620 | 1.240 | .921 | .580 | .263 | .388 |
| | | 4 | .375 | 1.049 | .620 | 1.490 | .979 | .654 | .302 | .427 |
| | | 5 | .380 | 1.194 | .680 | 1.495 | 1.009 | .726 | .320 | .445 |
| | | 6 | .380 | 1.303 | .750 | 1.610 | 1.079 | .798 | .369 | .494 |
| | | 8 | .438 | 1.465 | .880 | 1.722 | 1.332 | 1.014 | .445 | .632 |
| | | 10 | .515 | 1.680 | .967 | 2.037 | 1.488 | 1.158 | .501 | .688 |
| | | 12 | .570 | 1.727 | 1.122 | 2.368 | 1.667 | 1.447 | .588 | .775 |
| | | 16Z | .620 | 2.101 | 1.480 | 2.583 | 1.879 | 1.736 | .609 | .857 |
| | | 20Z | .640 | 2.255 | 1.750 | 2.946 | 2.058 | 2.328 | .697 | 1.017 |
| | | 24Z | .765 | 2.400 | 2.150 | 3.210 | 2.400 | 2.621 | .785 | 1.127 |

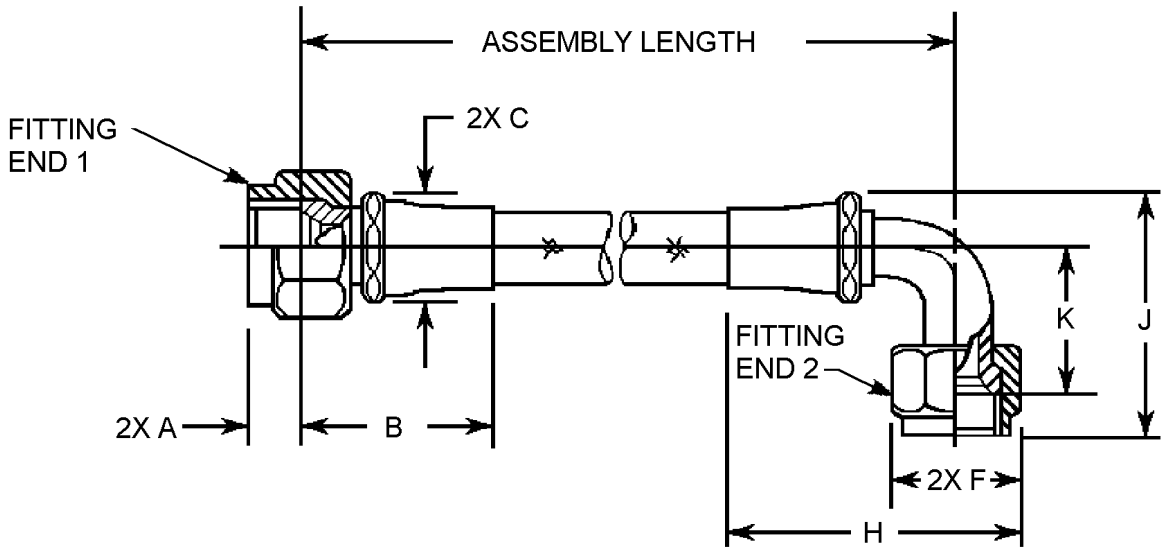
NOTES:
1. Style B uses class 1 fittings which are made from CRES.
2. Dimension “E (max)”is an installation dimension and not a design dimension.

FIGURE 2. Style B hose assembly dimensions.



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Hose assembly dimensions for style C: See figure 3.




| Fitting ends | | Hose | Dimensions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | size | A (ref) | B (max) | C (max) | F (max) | H (max) | J (max) | K (min) | K (max) |
| Straight | 90° | 3/4 | .372 | 1.049 | .620 | .580 | 1.845 | 1.320 | .510 | .635 |
| | | 4 | .375 | 1.049 | .620 | .654 | 1.885 | 1.370 | .560 | .685 |
| | | 5 | .380 | 1.194 | .680 | .726 | 1.980 | 1.542 | .635 | .822 |
| | | 6 | .380 | 1.303 | .750 | .798 | 2.205 | 1.640 | .700 | .885 |
| | | 8 | .438 | 1.465 | .880 | 1.014 | 2.495 | 1.938 | .810 | 1.060 |
| | | 10 | .515 | 1.680 | .967 | 1.158 | 2.955 | 2.340 | 1.091 | 1.341 |
| | | 12 | .570 | 1.727 | 1.122 | 1.447 | 3.705 | 2.722 | 1.341 | 1.591 |
| | | 16Z | .620 | 2.101 | 1.480 | 1.736 | 4.100 | 3.075 | 1.465 | 1.715 |
| | | 20Z | .640 | 2.255 | 1.750 | 2.328 | 4.835 | 3.637 | 1.747 | 2.122 |
| | | 24Z | .765 | 2.400 | 2.150 | 2.621 | 5.480 | 4.212 | 1.997 | 2.372 |

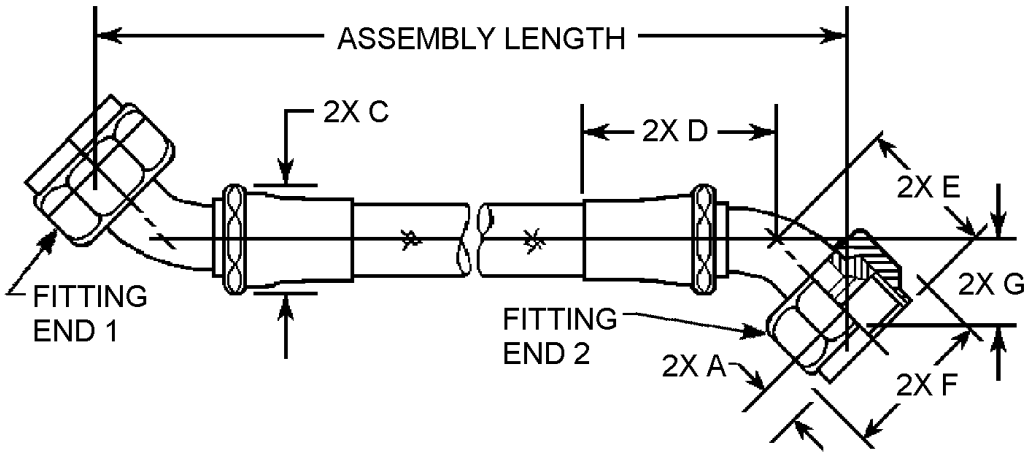
NOTES:
1. Style C uses class 1 fittings which are made from CRES.
2. Dimension “J (max)” is an installation dimension and not a design dimension.

FIGURE 3. Style C hose assembly dimensions.



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Hose assembly dimensions for style D: See figure 4.




| Fitting ends | | Hose | Dimensions | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | size | A (ref) | C (max) | D (max) | E (max) | F (max) | G (min) | G (max) |
| 45° | 45° | 3/4 | .372 | .620 | 1.240 | .921 | .580 | .263 | .388 |
| | | 4 | .375 | .620 | 1.490 | .979 | .654 | .302 | .427 |
| | | 5 | .380 | .680 | 1.495 | 1.009 | .726 | .320 | .445 |
| | | 6 | .380 | .750 | 1.610 | 1.079 | .798 | .369 | .494 |
| | | 8 | .438 | .880 | 1.722 | 1.332 | 1.014 | .445 | .632 |
| | | 10 | .515 | .967 | 2.037 | 1.488 | 1.158 | .501 | .688 |
| | | 12 | .570 | 1.122 | 2.368 | 1.667 | 1.447 | .588 | .775 |
| | | 16Z | .620 | 1.480 | 2.583 | 1.879 | 1.736 | .609 | .857 |
| | | 20Z | .640 | 1.750 | 2.946 | 2.058 | 2.328 | .697 | 1.017 |
| | | 24Z | .765 | 2.150 | 3.210 | 2.400 | 2.621 | .785 | 1.127 |

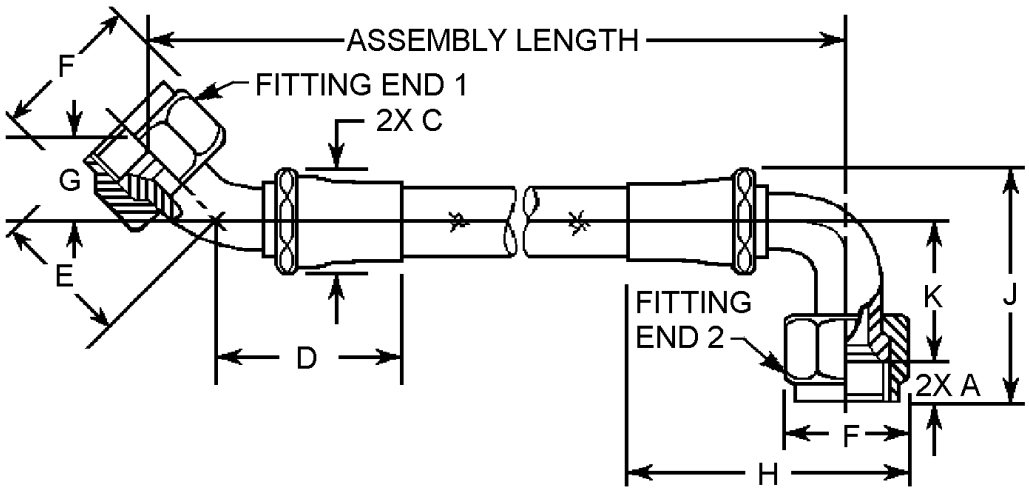
NOTES:
1. Style D uses class 1 fittings which are made from CRES.
2. Dimensions “E (max)” are installation dimensions and not design dimensions.

FIGURE 4. Style D hose assembly dimensions.



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Hose assembly dimensions for style E: See figure 5.




| Fitting ends | | Hose size | Dimensions | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | | A (ref) | C (max) | D (max) | E (max) | F (max) | G (min) | G (max) | H (max) | J (max) | K (min) | K (max) |
| 45° | 90° | 3/4 | .372 | .620 | 1.240 | .921 | .580 | .263 | .388 | 1.845 | 1.320 | .510 | .635 |
| | | 4 | .375 | .620 | 1.490 | .979 | .654 | .302 | .427 | 1.885 | 1.370 | .560 | .685 |
| | | 5 | .380 | .680 | 1.495 | 1.009 | .726 | .320 | .445 | 1.980 | 1.542 | .635 | .822 |
| | | 6 | .380 | .750 | 1.610 | 1.079 | .798 | .369 | .494 | 2.205 | 1.640 | .700 | .885 |
| | | 8 | .438 | .880 | 1.722 | 1.332 | 1.014 | .445 | .632 | 2.495 | 1.938 | .810 | 1.060 |
| | | 10 | .515 | .967 | 2.037 | 1.488 | 1.158 | .501 | .688 | 2.955 | 2.340 | 1.091 | 1.341 |
| | | 12 | .570 | 1.122 | 2.368 | 1.667 | 1.447 | .588 | .775 | 3.705 | 2.722 | 1.341 | 1.591 |
| | | 16Z | .620 | 1.480 | 2.583 | 1.879 | 1.736 | .609 | .857 | 4.100 | 3.075 | 1.465 | 1.715 |
| | | 20Z | .640 | 1.750 | 2.946 | 2.058 | 2.328 | .697 | 1.017 | 4.835 | 3.637 | 1.747 | 2.122 |
| | | 24Z | .765 | 2.150 | 3.210 | 2.400 | 2.621 | .785 | 1.127 | 5.480 | 4.212 | 1.997 | 2.372 |

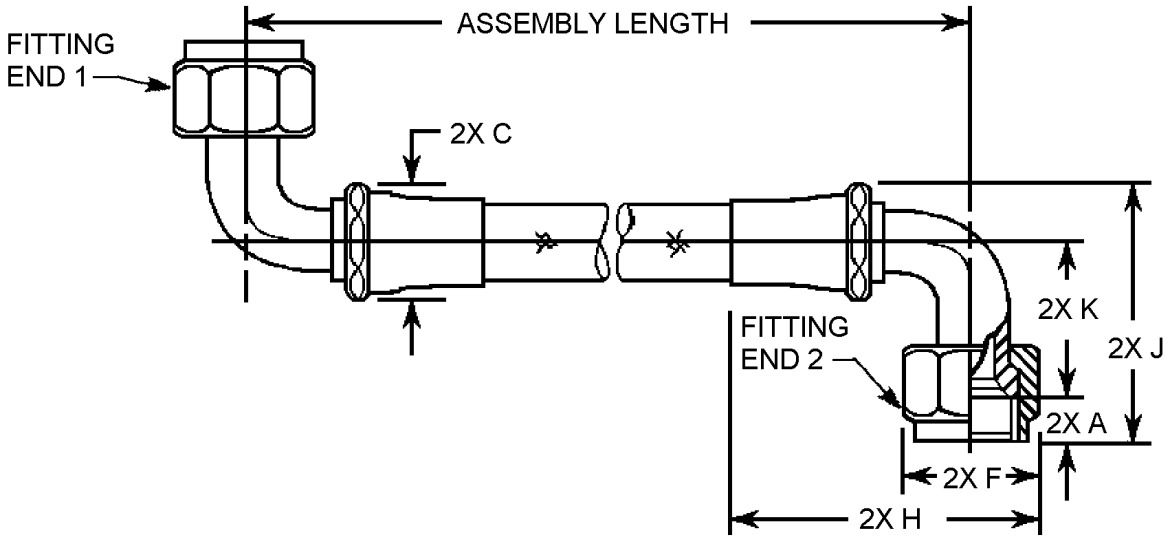
NOTES:
1. Style E uses class 1 fittings which are made from CRES.
2. Dimensions “E (max)” and “J (max)” are installation dimensions and not design dimensions.

FIGURE 5. Style E hose assembly dimensions.



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Hose assembly dimensions for style F: See figure 6.




| Fitting ends | | Hose | Dimensions | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | size | A (ref) | C (max) | F (max) | H (max) | J (max) | K (min) | K (max) |
| 90° | 90° | 3/4 | .372 | .620 | .580 | 1.845 | 1.320 | .510 | .635 |
| | | 4 | .375 | .620 | .654 | 1.885 | 1.370 | .560 | .685 |
| | | 5 | .380 | .680 | .726 | 1.980 | 1.542 | .635 | .822 |
| | | 6 | .380 | .750 | .798 | 2.205 | 1.640 | .700 | .885 |
| | | 8 | .438 | .880 | 1.014 | 2.495 | 1.938 | .810 | 1.060 |
| | | 10 | .515 | .967 | 1.158 | 2.955 | 2.340 | 1.091 | 1.341 |
| | | 12 | .570 | 1.122 | 1.447 | 3.705 | 2.722 | 1.341 | 1.591 |
| | | 16Z | .620 | 1.480 | 1.736 | 4.100 | 3.075 | 1.465 | 1.715 |
| | | 20Z | .640 | 1.750 | 2.328 | 4.835 | 3.637 | 1.747 | 2.122 |
| | | 24Z | .765 | 2.150 | 2.621 | 5.480 | 4.212 | 1.997 | 2.372 |

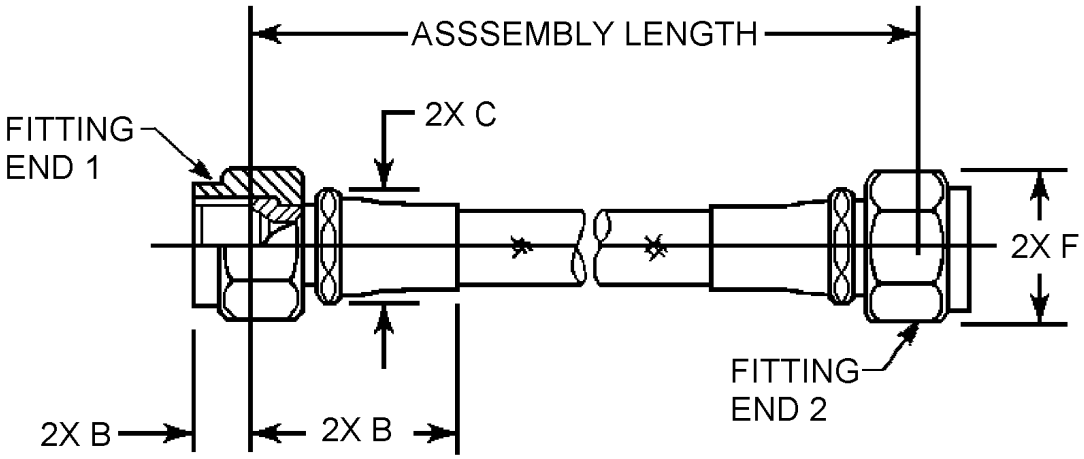
NOTES:
1. Style F uses class 1 fittings which are made from CRES.
2. Dimension “J (max)” is installation dimension and not a design dimension.

FIGURE 6. Style F hose assembly dimensions.



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Hose assembly dimensions for style G: See figure 7.




| Fitting ends | | Hose | Dimensions | | | |
|---|---|---|---|---|---|---|
| 1 | 2 | size | A (ref) | B ( max) | C (max) | F (max) |
| Straight | Straight | 8 | .438 | 1.465 | .880 | 1.014 |
| | | 10 | .515 | 1.680 | .967 | 1.158 |
| | | 12 | .570 | 1.727 | 1.122 | 1.447 |
| | | 16Z | .620 | 2.101 | 1.480 | 1.736 |
| | | 20Z | .640 | 2.255 | 1.750 | 2.328 |
| | | 24Z | .765 | 2.400 | 2.150 | 2.621 |

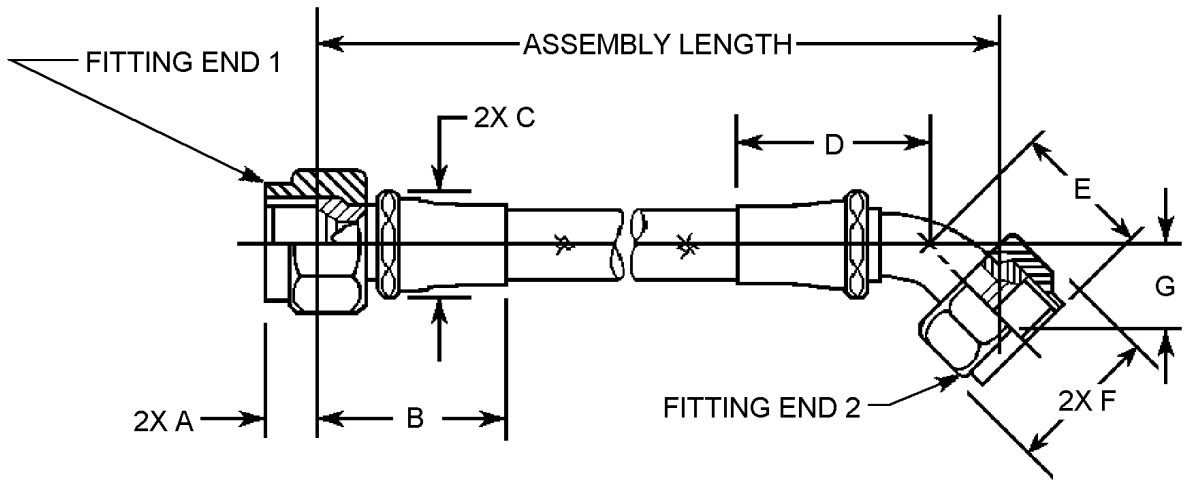
NOTE: Style G uses class 2 fittings which are made from a combination of aluminum and CRES.

FIGURE 7. Style G hose assembly dimensions.



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Hose assembly dimensions for style H: See figure 8.




| Fitting ends | | Hose size | Dimensions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | | A ref) | B (max) | C (max) | D (max) | E (max) | F (max) | G (min) | G (max) |
| Straight | 45° | 8 | .438 | 1.465 | .880 | 1.722 | 1.332 | 1.014 | .445 | .632 |
| | | 10 | .515 | 1.680 | .967 | 2.037 | 1.488 | 1.158 | .501 | .688 |
| | | 12 | .570 | 1.727 | 1.122 | 2.368 | 1.667 | 1.447 | .588 | .775 |
| | | 16Z | .620 | 2.101 | 1.480 | 2.583 | 1.879 | 1.736 | .609 | .857 |
| | | 20Z | .640 | 2.255 | 1.750 | 2.946 | 2.058 | 2.328 | .697 | 1.017 |
| | | 24Z | .765 | 2.400 | 2.150 | 3.210 | 2.400 | 2.621 | .785 | 1.127 |

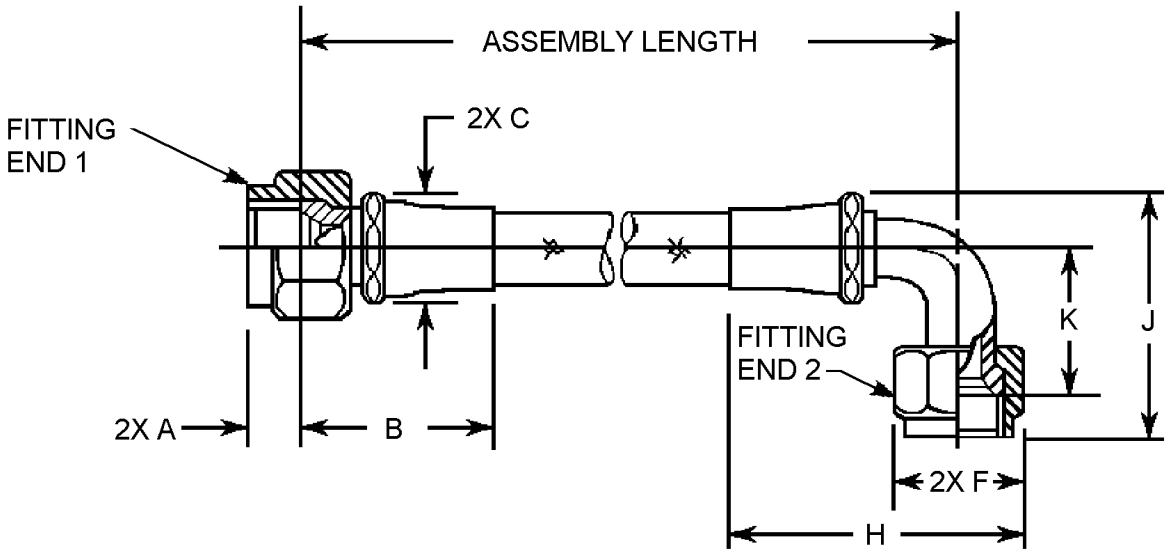
NOTES:
1. Style H uses class 2 fittings which are made from a combination of aluminum and CRES.
2. Dimension “E (max)”is an installation dimension and not a design dimension.

FIGURE 8. Style H hose assembly dimensions.



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Hose assembly dimensions for style J: See figure 9.




| Fitting ends | | Hose size | Dimensions | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | | A (ref) | B (max) | C (max) | F (max) | H (max) | J (max) | K (min) | K (max) |
| Straight | 90° | 8 | .438 | 1.465 | .880 | 1.014 | 2.495 | 1.938 | .810 | 1.060 |
| | | 10 | .515 | 1.680 | .967 | 1.158 | 2.955 | 2.340 | 1.091 | 1.341 |
| | | 12 | .570 | 1.727 | 1.122 | 1.447 | 3.705 | 2.722 | 1.341 | 1.591 |
| | | 16Z | .620 | 2.101 | 1.480 | 1.736 | 4.100 | 3.075 | 1.465 | 1.715 |
| | | 20Z | .640 | 2.255 | 1.750 | 2.328 | 4.835 | 3.637 | 1.747 | 2.122 |
| | | 24Z | .765 | 2.400 | 2.150 | 2.621 | 5.480 | 4.212 | 1.997 | 2.372 |

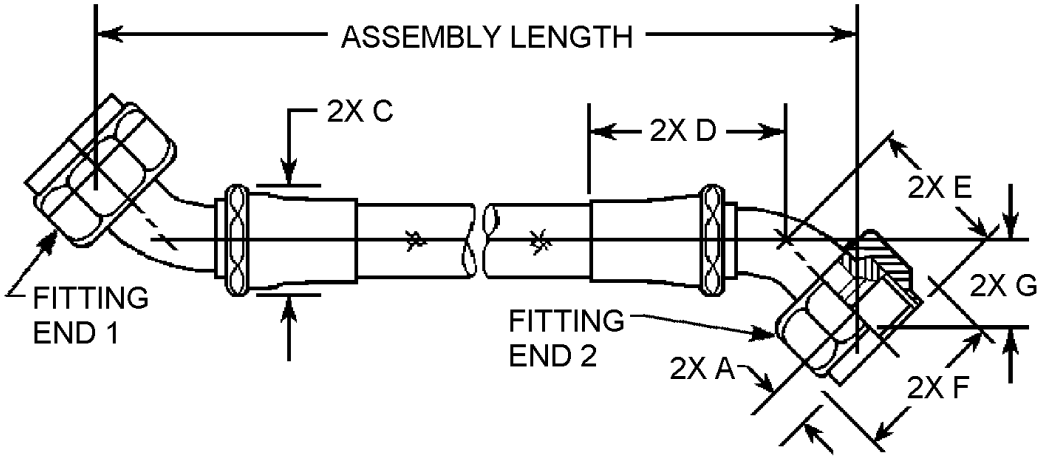
NOTES:
1. Style J uses class 2 fittings which are made from a combination of aluminum and CRES.
2. Dimension “J (max)” is an installation dimension and not a design dimension.

FIGURE 9. Style J hose assembly dimensions.



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Hose assembly dimensions for style K: See figure 10.




| Fitting ends | | Hose size | Dimensions | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | | A (ref) | C (max) | D (max) | E (max) | F (max) | G (min) | G (max) |
| 45° | 45° | 8 | .438 | .880 | 1.722 | 1.332 | 1.014 | .445 | .632 |
| | | 10 | .515 | .967 | 2.037 | 1.488 | 1.158 | .501 | .688 |
| | | 12 | .570 | 1.122 | 2.368 | 1.667 | 1.447 | .588 | .775 |
| | | 16Z | .620 | 1.480 | 2.583 | 1.879 | 1.736 | .609 | .857 |
| | | 20Z | .640 | 1.750 | 2.946 | 2.058 | 2.328 | .697 | 1.017 |
| | | 24Z | .765 | 2.150 | 3.210 | 2.400 | 2.621 | .785 | 1.127 |

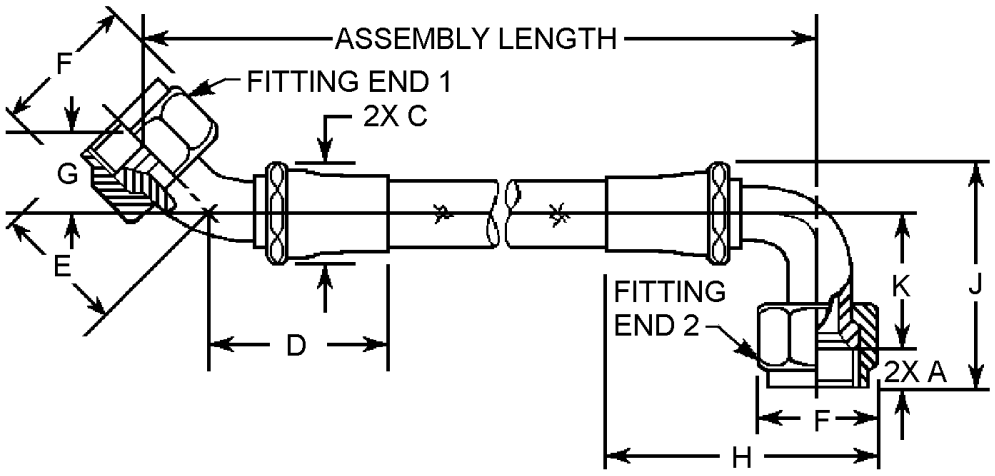
NOTES:
1. Style K uses class 2 fittings which are made from a combination of aluminum and CRES.
2. Dimension “E (max)” is an installation dimension and not a design dimension.

FIGURE 10. Style K hose assembly dimensions.



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Hose assembly dimensions for style M: See figure 11.




| Fitting ends | | Hose size | Dimensions | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | | A (ref) | C (max) | D (max) | E (max) | F (max) | G (min) | G (max) | H (max) | J (max) | K (min) | K (max) |
| 45° | 90° | 8 | .438 | .880 | 1.722 | 1.332 | 1.014 | .445 | .632 | 2.495 | 1.938 | .810 | 1.060 |
| | | 10 | .515 | .967 | 2.037 | 1.488 | 1.158 | .501 | .688 | 2.955 | 2.340 | 1.091 | 1.341 |
| | | 12 | .570 | 1.122 | 2.368 | 1.667 | 1.447 | .588 | .775 | 3.705 | 2.722 | 1.341 | 1.591 |
| | | 16Z | .620 | 1.480 | 2.583 | 1.879 | 1.736 | .609 | .857 | 4.100 | 3.075 | 1.465 | 1.715 |
| | | 20Z | .640 | 1.750 | 2.946 | 2.058 | 2.328 | .697 | 1.017 | 4.835 | 3.637 | 1.747 | 2.122 |
| | | 24Z | .765 | 2.150 | 3.210 | 2.400 | 2.621 | .785 | 1.127 | 5.480 | 4.212 | 1.997 | 2.372 |

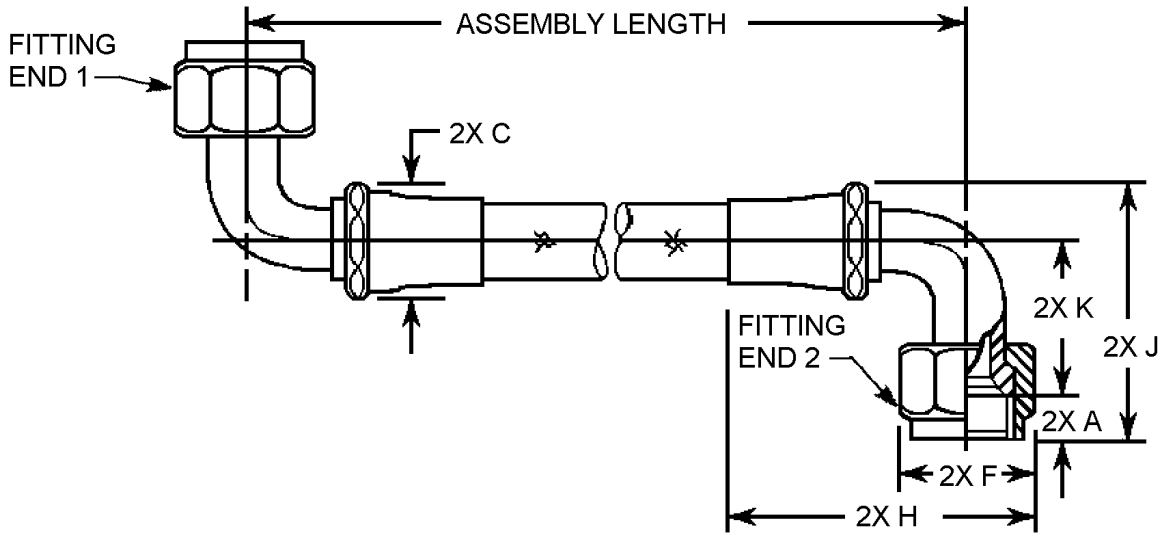
NOTES:
1. Style M uses class 2 fittings which are made from a combination of aluminum and CRES.
2. Dimensions “E (max)” and “J (max)” are installation dimensions and not design dimensions.

FIGURE 11. Style M hose assembly dimensions.



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Hose assembly dimensions for style N: See figure 12.




| Fitting ends | | Hose | Dimensions | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | size | A (ref) | C (max) | F (max) | H (max) | J (max) | K (min) | K (max) |
| 90° | 90° | 8 | .438 | .880 | 1.014 | 2.495 | 1.938 | .810 | 1.060 |
| | | 10 | .515 | .967 | 1.158 | 2.955 | 2.340 | 1.091 | 1.341 |
| | | 12 | .570 | 1.122 | 1.447 | 3.705 | 2.722 | 1.341 | 1.591 |
| | | 16Z | .620 | 1.480 | 1.736 | 4.100 | 3.075 | 1.465 | 1.715 |
| | | 20Z | .640 | 1.750 | 2.328 | 4.835 | 3.637 | 1.747 | 2.122 |
| | | 24Z | .765 | 2.150 | 2.621 | 5.480 | 4.212 | 1.997 | 2.372 |

NOTES:
1 Style N uses class 2 fittings which are made from a combination of aluminum and CRES.
2. Dimension “J (max)” is an installation dimension and not a design dimension.

FIGURE 12. Style N hose assembly dimensions.



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REQUIREMENTS:

Hose shall be in accordance with MIL-DTL-32435.

Hose assembly size code: See table I.

TABLE I. Hose assembly size code.

| Size | Reference tube OD | Size code |
|---|---|---|
| 3/4 | 1/ | B |
| 4 | .250 | E |
| 5 | .313 | F |
| 6 | .375 | G |
| 8 | .500 | H |
| 10 | .625 | J |
| 12 | .750 | K |
| 16Z | 1.000 | M |
| 20Z | 1.250 | N |
| 24Z | 1.500 | P |

1/ Size 3/4 hose assembly consists of a size 4 tube OD hose assembled with size 3 tube OD fittings.

Fittings: The swivel nut and cone seat shall mate with SAE-AS33514 fitting. The swivel nut threads shall be in accordance with SAE-AS8879.

Elbow fittings: The ovality of the circular cross section within the angle of the bend of the end fitting elbow shall not exceed 7.5% of the nominal tubing OD, see table II.

TABLE II. Elbow fitting ovality. 1/

| Assembly size | 3/4 | 4 | 5 | 6 | 8 | 10 | 12 | 16 | 20 | 24 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ball diameter | .080 | .119 | .174 | .230 | .313 | .391 | .515 | .750 | .969 | 1.188 |

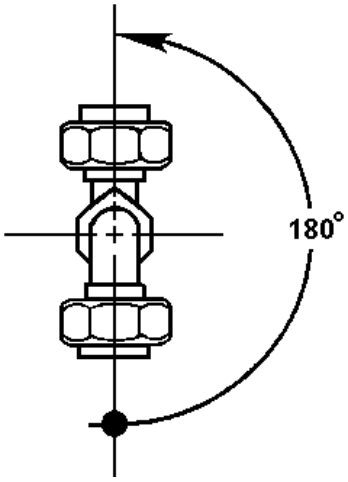
1/ The fitting elbow bend shall be examined for ovality by rolling a ball of applicable diameter through the elbow fitting. The ball shall pass freely through the fitting elbow bend.

Assembly classification: Class 1 and class 2 hose assemblies, as specified in MIL-DTL-32434, have been incorporated into the Part or Identifying Number (PIN) as a part of styles.

Angular alignment: Hose assemblies with elbow fittings on each end shall have the angular orientation between the elbows measured counter-clockwise from the centerline of the nearest fitting, positioned at six-o'clock, to the centerline of the other fitting (see figure 13). When applicable, the angular alignment shall be expressed in three digits and specified in the PIN.



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NOTE: Angular alignment shall be measured in degrees with a tolerance of ±2°.

FIGURE 13. Measurement of angular alignment between elbow fittings.

Protective sleeve: If required, the hose assembly shall include a protective sleeve (see table III) and its code shall be included in the PIN. Fire protective sleeve shall be subjected to testing in accordance with MIL-DTL-32434.

TABLE III. Protective sleeve code.

| Code | Type |
|---|---|
| Blank | None |
| A | SAE-AS1072 sleeve, fire protection, silicone covered, temperature ranging from -65°F to 450°F and intermittently to 500°F, secured with CRES bands as required. 1/ |
| B | SAE-AS1073 - code B sleeve, abrasion protection, heat shrinkable, black polyolefin, temperature ranging from -65°F to 250°F. |
| C | SAE-AS1291 - code A sleeve, chafe guard, extruded seamless white PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required. |
| D | SAE-AS1291 - code C sleeve, chafe guard, extruded seamless transparent FEP, temperature ranging from -65°F to 350°F, secured with CRES bands as required. |
| E | SAE-AS1298 sleeve, heavy wall chafe guard, extruded seamless black PTFE, temperature ranging from -65°F to 450°F, secured with CRES bands as required. |
| L | Lock-wire hole |
| F | Code A + L |
| G | Code B + L |
| H | Code C + L |
| J | Code D + L |
| K | Code E + L |

1/ To prevent wicking of fluids, the cut end of the fire protective sleeve (code A) shall be coated with room temperature vulcanized (RTV) silicone rubber prior to installation. After installation, cracks and voids in the fire protective sleeve shall be coated with RTV rubber to prevent exposure of fiberglass.

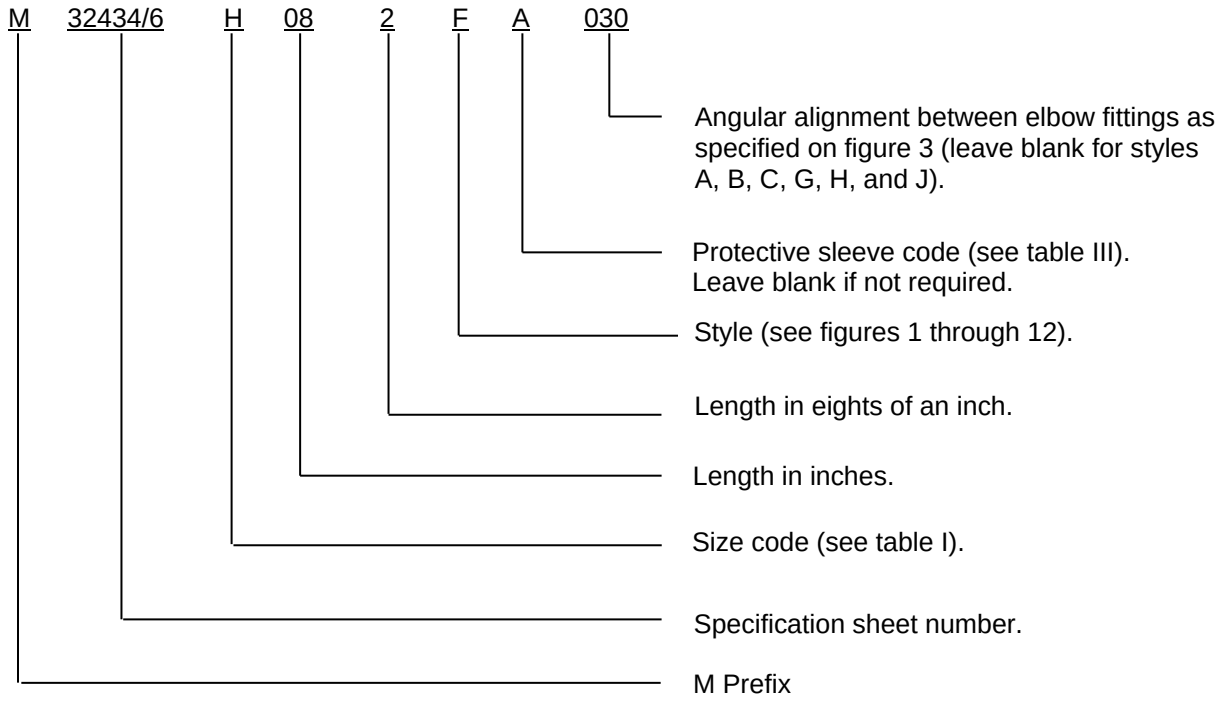


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Assembly length and tolerance: Hose assembly shall be furnished in lengths as specified in the contract or purchase order (see figures 1 through 12); however, tolerances on the length of each hose assembly shall be as follows:

a. ±1/8 inch for lengths under 18 inches.
b. ±1/4 inch for lengths from 18 inches to 36 inches.
c. ±1/2 inch for lengths from 36 inches to 50 inches.
d. ±1% for lengths over 50 inches.

PIN: The PIN consists of letter M, specification sheet number, hose size code, hose length in inches and an eighths of an inch, hose assembly style, sleeve code, and angular alignment in degree between elbow fittings.




Example: M32434/6H082FA030 represents 1/2inch OD PTFE convoluted hose assembly, 8.25 inches long, flare-to-flare permanently attached fittings with fire protective sleeve and 30° angular alignment between elbows.

Referenced documents. In addition to MIL-DTL-32434, this document references the following:

| | |
|---|---|
| MIL-DTL-32435 | SAE-AS1291 |
| SAE-AS1072 | SAE-AS8879 |
| SAE-AS1073 | SAE-AS33514 |
| SAE-AS1298 | |



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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4720-2012-024)

Review activities:
Army - AR, AT, EA, MI
Navy - MC, SA, SH
Air Force - 70

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